

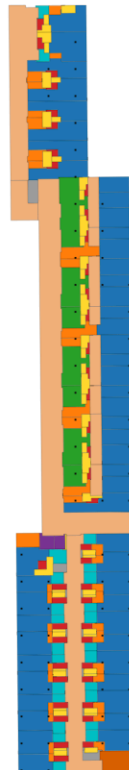
Workflow: MSD Site 530 Building 1214 BGR Graph Prediction

For this assignment, I selected one building from the Modified Swiss Dwellings dataset to generate a simplified BGR graph model and compare a manual label with the prediction from the pretrained model.

The selected case was:

MSD site 530 / building 1214 / floor 11739 - internal corridors by unit_id

ROOM BATHROOM KITCHEN LIVING_DINING STOREROOM INTERIOR_CORRIDOR CORRIDOR_COMMON SHAFT ELEVATOR STAIRCASE DERIVED_GRID_COLUMN



The original dataset stores the geometry as WKT polygon text, so I did not import the CSV directly into Rhino. Instead, I filtered the dataset in Python using the selected IDs and extracted only the useful area entities. Walls, doors, windows, entrance doors, openings, and separators were excluded.

clean_layer	entity_subtype	area_id	unit_id	exported	polygon_parts	original_area	offset_trimmed_area
1	core	CORRIDOR	669436.0	True	1	156.867257	205.285794
2	core	CORRIDOR	579130.0	True	1	63.968494	94.581258
3	core	CORRIDOR	669435.0	True	1	57.634797	81.671046
4	core	CORRIDOR	579073.0	True	1	14.102048	18.807957
5	core	STAIRCASE	669434.0	True	1	11.0643	15.687645
6	core	CORRIDOR	916963.0	True	1	6.792488	12.767298
7	core	CORRIDOR	916955.0	True	1	6.67976	12.640662
8	core	CORRIDOR	916959.0	True	1	6.673074	12.654188
9	core	CORRIDOR	916951.0	True	1	6.430749	12.353462
10	core	CORRIDOR	916964.0	True	1	4.567924	8.041397
11	core	CORRIDOR	916960.0	True	1	4.420564	7.133711
12	core	CORRIDOR	916969.0 59436.0	True	1	4.334219	6.471044
13	core	ELEVATOR	578936.0	True	1	2.620216	4.28289
14	core	CORRIDOR	916952.0	True	1	2.479838	4.579147
15	core	CORRIDOR	916956.0	True	1	2.457756	4.545088
16	core	ELEVATOR	578905.0	True	1	2.27239	2.713652
17	core	CORRIDOR	917002.0 59446.0	True	1	1.873332	3.371122
18	core	CORRIDOR	916994.0 59448.0	True	1	1.873332	3.371122
19	core	CORRIDOR	916986.0 59440.0	True	1	1.846321	3.327341
20	core	CORRIDOR	916981.0 59435.0	True	1	1.836269	3.309693
21	core	CORRIDOR	916973.0 59431.0	True	1	1.835625	3.309049
22	core	CORRIDOR	916978.0 59437.0	True	1	1.829188	3.323235
23	core	CORRIDOR	917005.0 59439.0	True	1	1.820582	2.464282
24	core	CORRIDOR	916980.0 59454.0	True	1	1.803636	2.448032
25	core	CORRIDOR	916972.0 59444.0	True	1	1.770805	3.323732
26	core	CORRIDOR	917004.0 59431.0	True	1	1.774197	3.230925
27	core	CORRIDOR	916971.0 59427.0	True	1	1.741886	2.372851
28	core	CORRIDOR	916988.0 59443.0	True	1	1.735668	2.369581
29	core	CORRIDOR	917003.0 59438.0	True	1	1.735373	2.367864
30	core	CORRIDOR	916996.0 59451.0	True	1	1.734428	2.368261
31	core	CORRIDOR	916989.0 59451.0	True	1	1.730689	3.13457
32	core	CORRIDOR	916966.0 59406.0	True	1	1.724564	2.799153
33	core	CORRIDOR	916997.0 59452.0	True	1	1.716086	3.133671
34	core	CORRIDOR	916979.0 59449.0	True	1	1.700333	2.326325
35	core	CORRIDOR	917009.0 59450.0	True	1	1.697997	3.180817
36	core	CORRIDOR	916995.0 59447.0	True	1	1.690287	2.310075
37	core	CORRIDOR	916987.0 59445.0	True	1	1.624781	2.234006
38	core	CORRIDOR	916941.0 59410.0	True	1	1.581143	3.101777
39	core	CORRIDOR	916944.0 59414.0	True	1	1.563568	3.075519
40	core	SHAFT	578967.0	True	1	1.560423	3.640171

The extracted spaces were mapped into BGR modeling categories. Rooms, bathrooms, kitchens, living/dining areas, and storerooms were assigned to offices/building. Corridors, elevators, staircases, shafts, and voids were assigned to core. Since the MSD floor did not include structural columns, I generated an auxiliary column grid inside the building perimeter and outside the core areas.

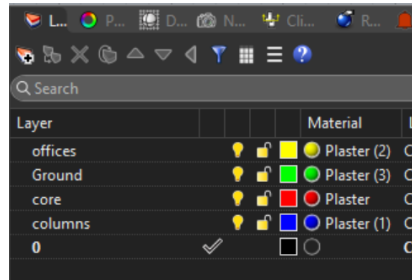
MSD_site530_building1214_floor11739_clean_spaces_union_summary.csv			
	clean_layer	entity_subtype	source_area_count
1	offices	BATHROOM	43
2	offices	KITCHEN	38
3	offices	LIVING_DINING	7
4	offices	ROOM	51
5	offices	STOREROOM	14
6	core	CORRIDOR	36
7	core	ELEVATOR	2
8	core	SHAFT	55
9	core	STAIRCASE	1

To prepare a clean Rhino reference, I applied a small offset to the individual space boundaries so the wall gaps were absorbed by adjacent spaces. Each space kept its own closed boundary, but all spaces were adjusted to form one connected network with no internal gaps. The final reference was exported as a DXF with the BGR layers:

- offices
- core
- Columns

In Rhino, I imported the DXF and used the closed boundaries to create planar surfaces. These surfaces were extruded into simplified 3D solids. I modeled the building using separate BGR layers:

- ground
- columns
- offices
- Core



Before exporting, I cleaned the Rhino model by checking for bad objects, duplicate geometry, open polysurfaces, naked edges, non-manifold edges, degenerate faces, and very small accidental surfaces. The goal was to keep only valid closed solid polysurfaces in each layer. Each layer was then exported separately as OBJ:

- ground.obj
- columns.obj
- offices.obj
- core.obj